



Group number: 4



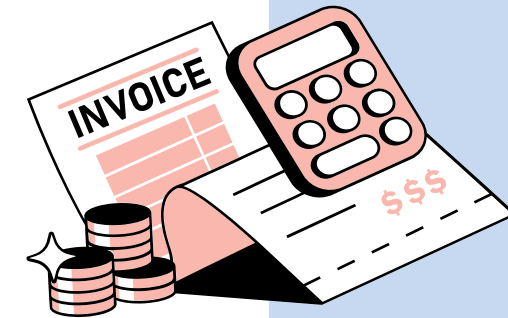
A Hierarchical Analysis of Structural Borrowing Costs in Texas Municipal Debt

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EXECUTIVE SUMMARY

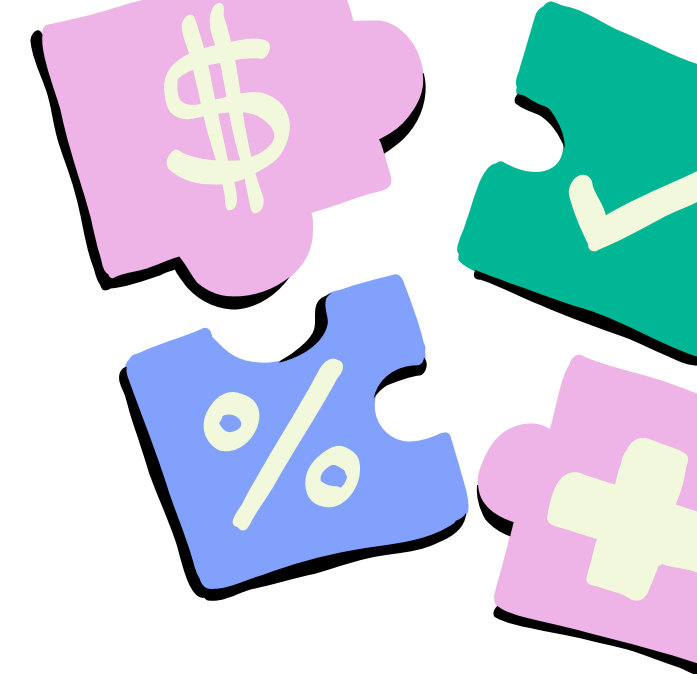
- **Objective:** Identify how pledge type & entity drive Texas debt costs (IPR)
- **Data:** 17,000+ debt issuances analyzed
- **Security Tax:** Revenue bonds carry a premium over GO bonds
- **Institutional DNA:** Government type is a key driver of borrowing cost differences
- **Market Saturation:** Large loans lose scale efficiency (non-linear effect)
- **Conclusion:** Borrowing cost driven by structural risk, not market averages





Introduction: Context & Purpose

- **The Context:** Municipal debt is the primary funding source for Texas infrastructure (schools, utilities, roads).
- **The Purpose:** To move beyond simple market averages and analyze the **structural drivers**—legal guarantees and entity identity—that dictate a loan's true lifecycle cost.
- **The Why:** Understanding these factors allows for better financial planning and risk management for local governments.



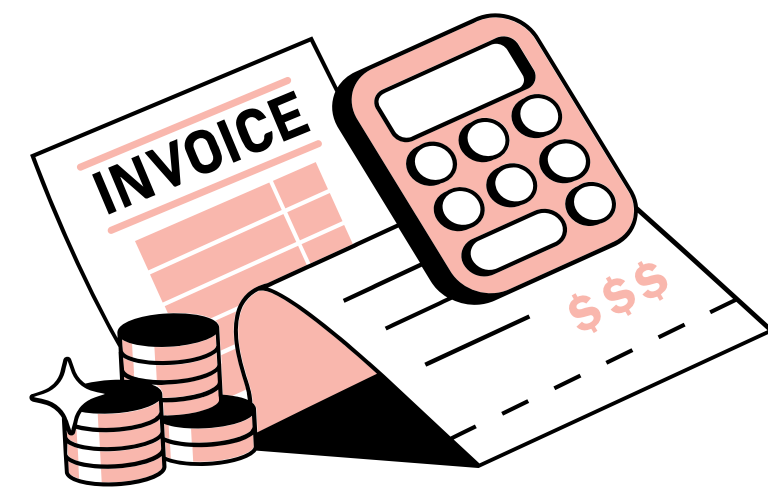
Problem Statement

- **Primary Question:** How does **Pledge Type** (GO vs. REV vs. LP) impact the long-term interest burden (IPR) when nested within specific **Government Categories**?
- **Sub-Questions:**
 1. Does a statistically significant "**Revenue Premium**" exist?
 2. How much cost is pre-determined by **Institutional Identity** (Government Type)?
 3. Is the relationship between loan size and cost linear or complex?



→ Dataset Overview

- Dataset Overview & Challenges
- The Data: ~17,400 unique Texas municipal debt issuances with 12 observations.
- Scale: Includes 7 distinct entity types (Cities, ISDs, Counties, Water Departments, etc.).
- Key Challenges:
 1. Data Integrity: Identifying and removing "Zombie Debt" (records with zero principal).
 2. Pseudo-replication: Cleaning duplicates to ensure each row represents a unique financial event.



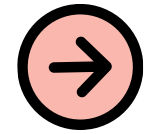
→ Core Metric: Interest-to-Principal Ratio (IPR)

- How (Calculation):

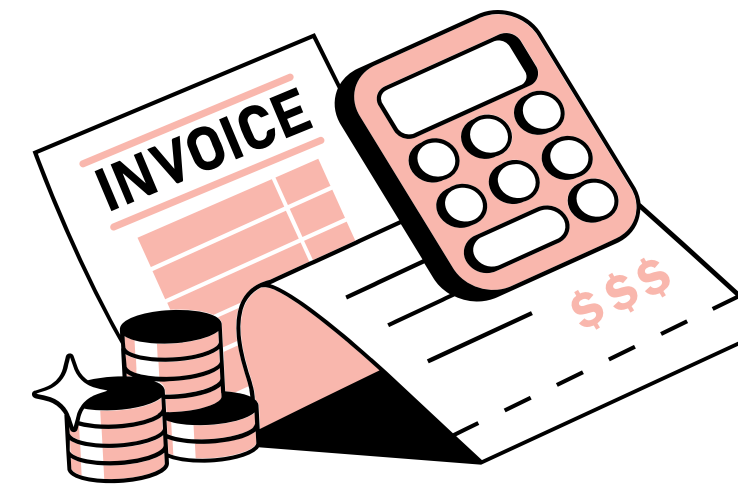
$$\text{IPR} = \frac{\text{Total Interest Outstanding}}{\text{Total Principal Outstanding}}$$

- Why (Normalization): It provides a standardized measure of borrowing cost.
- The Relation: IPR serves as the **Dependent Variable**.





Data Cleaning and Preprocessing



Raw Data
Ingestion
(~21,000
rows)

Outlier
Mitigation
(Filtering
invalid cases
where $IPR > 2.0$)

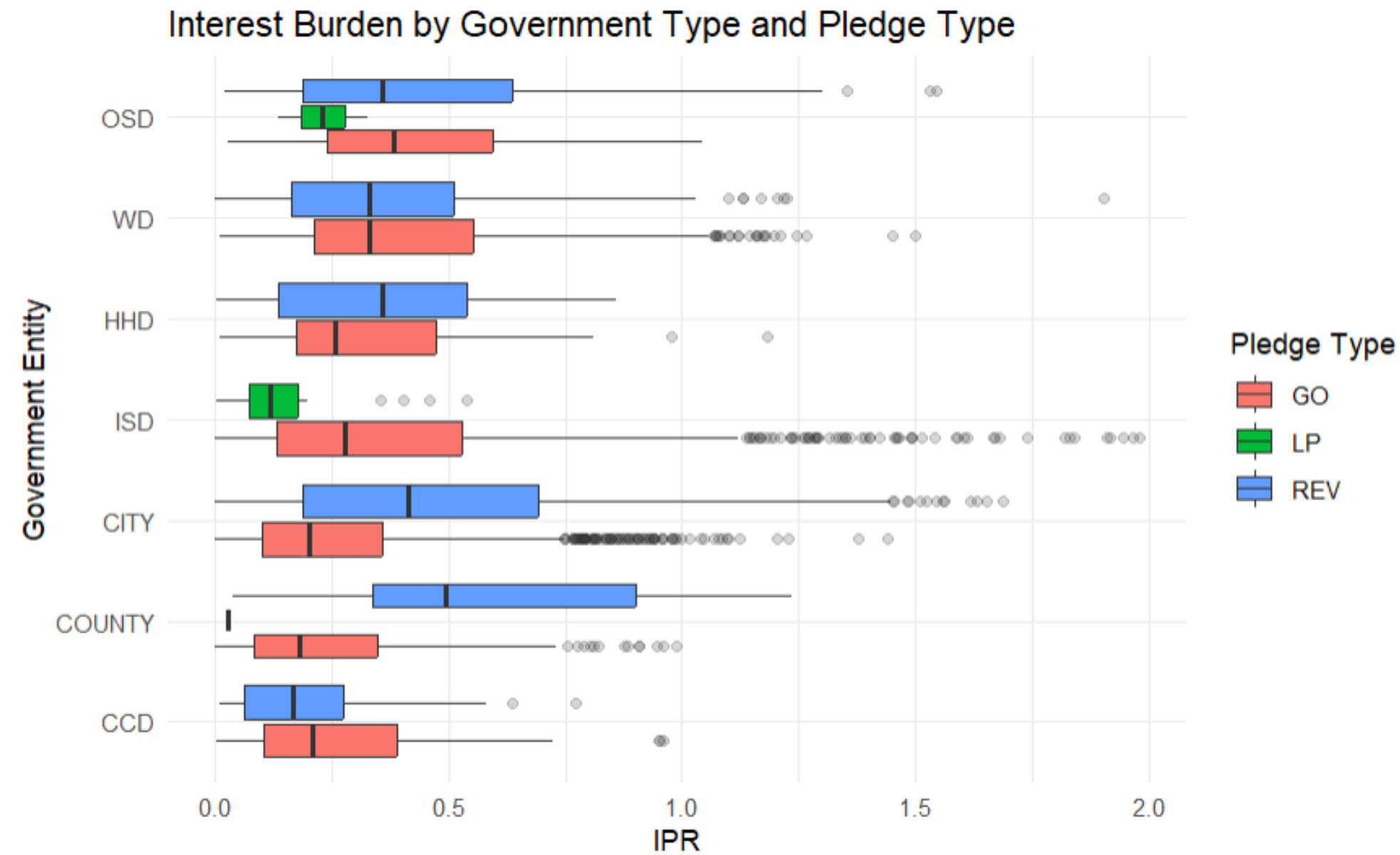
Log
Transformation
(Normalizing
Principal size to
stabilize variance)

Structural Audit
(Eliminating 0
Principal and 0
Interest entries)

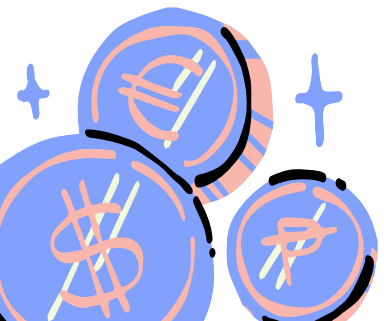
Deduplication
(Resolving
pseudo-
replication of
duplicate
issuances)

Temporal
Encoding
(distinguish
between
structural risk
and
macroeconomic
trends)

EDA: Visualizing the “Revenue Premium”



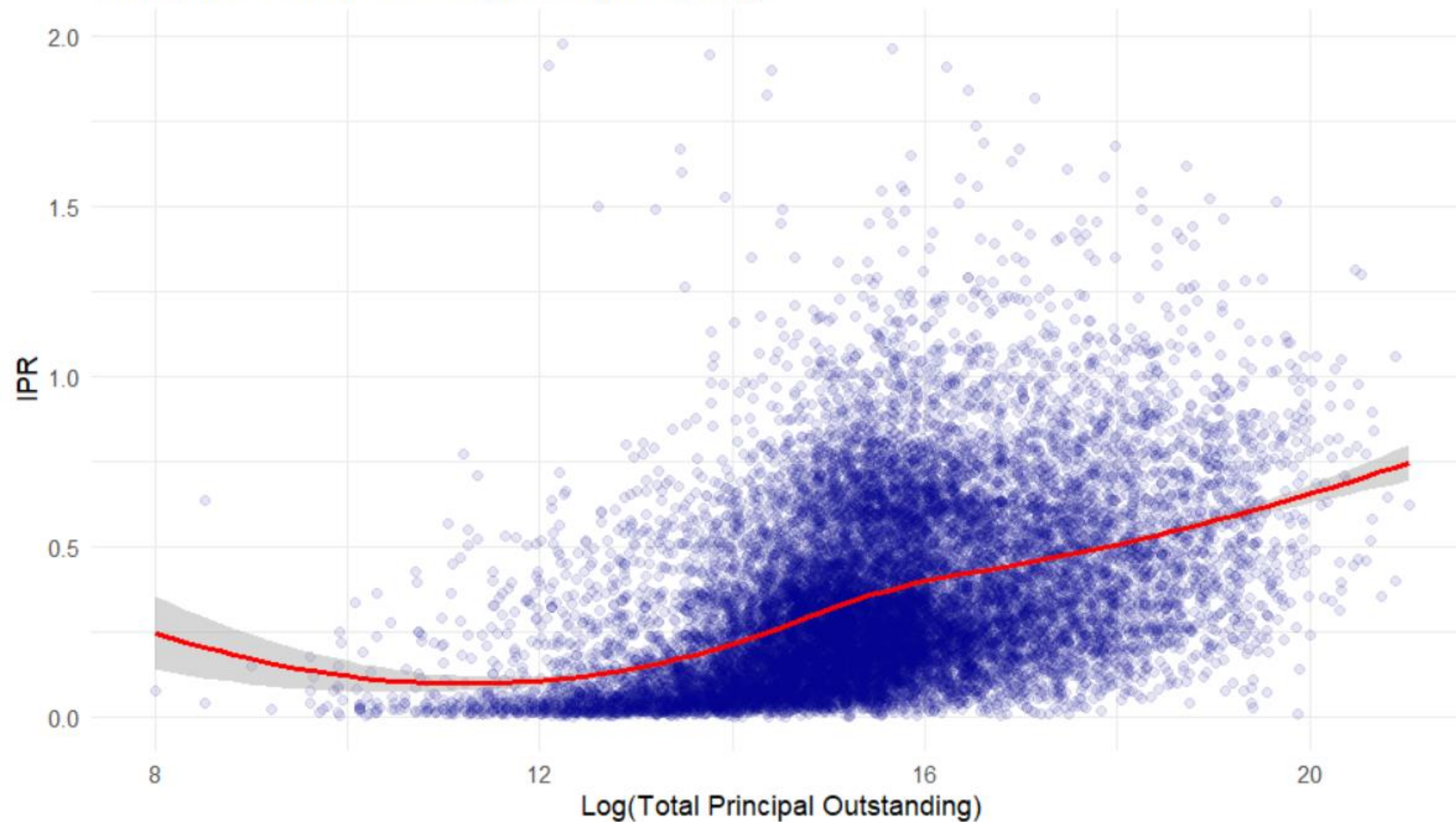
- **Observation**
Revenue bonds (REV) consistently show higher median IPR than GO bonds across most government types.
- **Key Insight:** Pledge type clearly influences borrowing cost -> this is not random variation.
- **Structural Pattern :** Different entity types behave differently, which supports using a hierarchical model



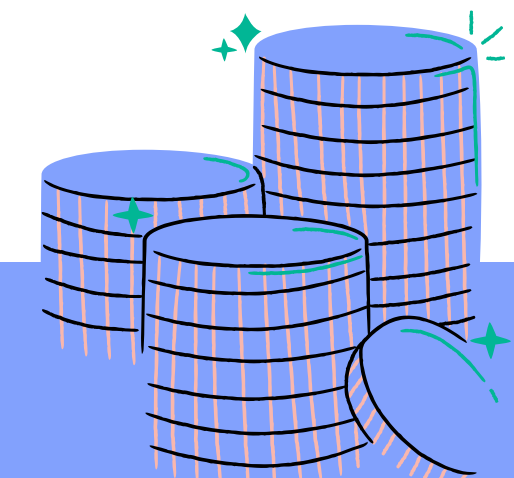
EDA: Testing Economies of Scale

Interest Burden vs. Issuance Size (Log Scale)

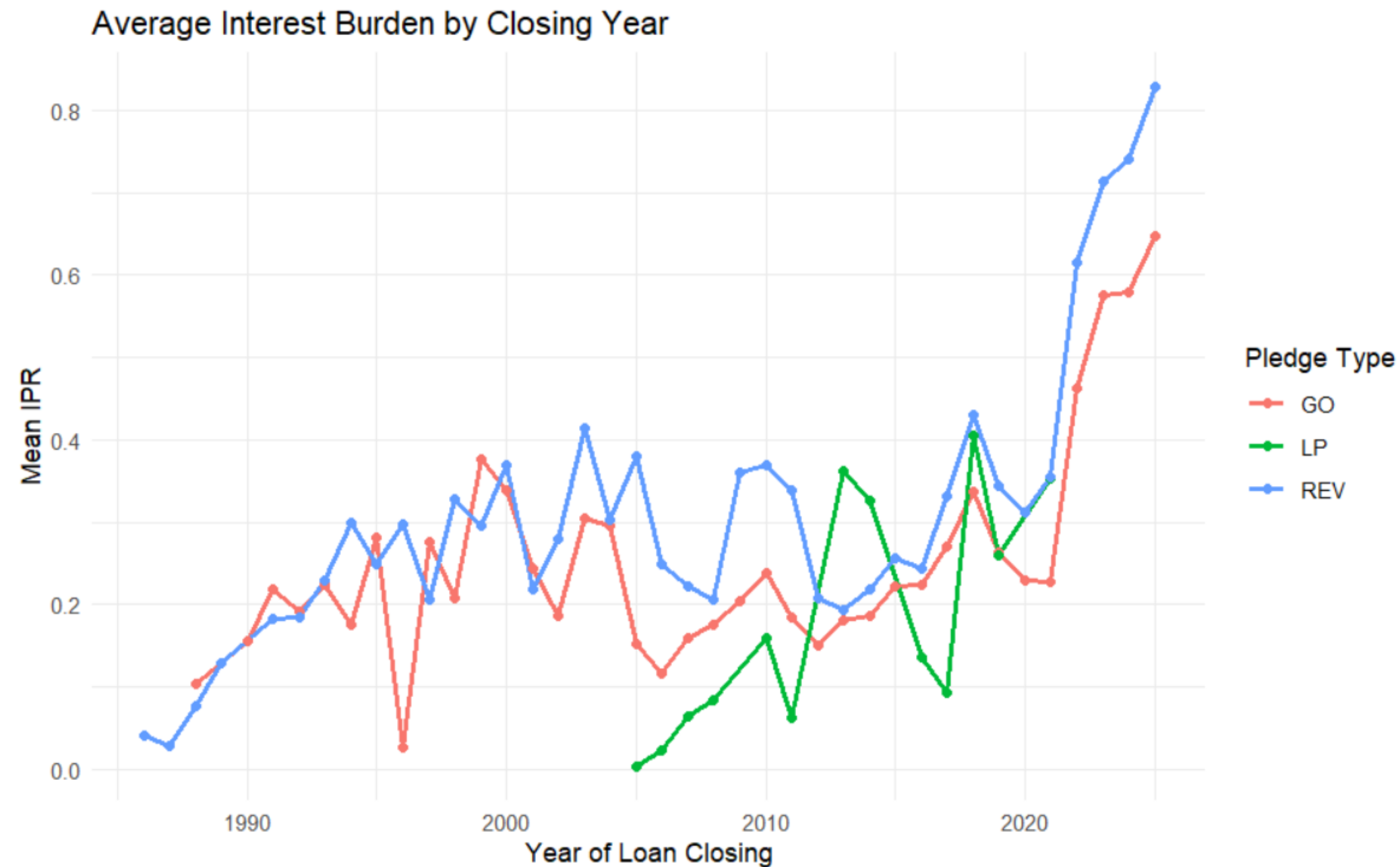
Testing for Economies of Scale in Municipal Borrowing



- **Observation**
The relationship between loan size (log principal) and IPR is **non-linear**.
- **Economies of Scale**
IPR declines with size initially, indicating **cost efficiencies for larger issuances**.
- **Diminishing Returns**
Beyond a point, IPR starts rising again, very large loans lose that cost advantage.
- **Modeling Note:**
A simple linear term is not enough, we need to account for this non-linearity.



EDA: Temporal Trends in Interest Burden



- **Trend Shift**
IPR remains stable before 2020, then jumps sharply after 2021.
- **Pledge Gap**
Revenue bonds remain more expensive than GO bonds throughout, the spread is consistent.
- **Volatility Pattern**
Lease Purchase (LP) bonds show **higher variability** due to smaller sample size.
- **Current Regime**
Recent years (2022–2025) reflect **elevated borrowing costs** relative to historical norms.





Model Structure & Integrity

Model Equation:

```
ipr ~ pledge_type + issue_purpose + log_principal + (1 | government_type)
```

All results are interpreted after controlling for loan size and issue purpose.

Multicollinearity Diagnostic (GVIF):

- All GVIF values are near 1.0, well below the threshold of 2.0
 - Pledge Type, Purpose, and Size each provide independent information.
 - The Revenue Premium is a real effect, not an overlap artifact.
- 

Model Integrity & Estimation Strategy

The Null Model (ICC)

- Intercept-only model used to estimate **intra-class correlation (ICC)**.
- About 6.2% variation comes from govt. type

The OLS Baseline:

- Simple regression without grouping
- Uses as a benchmark model

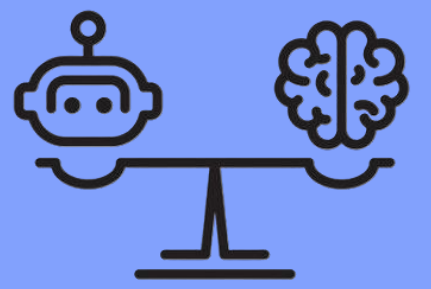
Hierarchical Linear Model (HLM)

- Accounts for clustering (loans within govt type)
- Allows each govt type to have its own baseline

Model Selection (AIC/ BIC)

- HLM achieves **lower AIC/BIC**, indicating superior model fit relative to OLS.

Model Comparison & Selection



AIC & BIC: The Statistical "Weigh-In"

To determine the superior model, we compared the standard OLS regression against our Hierarchical Linear Model (HLM) using information criteria:

Metric	OLS Model	HLM Model
AIC	-4,566.1	-4,791.9
BIC	-4,387.6	-4,605.6

The Verdict:

In model selection theory, a drop in $AIC > 10$ is considered definitive evidence of a superior model. Our drop of ~225 proves the HLM is a significantly better fit for the data.

Variance Explained (R^2)

- Marginal $R^2 = 0.324$ (fixed effects)
- Conditional $R^2 = 0.359$ (fixed + random effects)
- Indicates **meaningful contribution of group-level variation**

Likelihood Ratio Test (LRT)

$$p < 2.2 \times 10^{-16}$$

Rejects OLS in favor of hierarchical specification





Methodological Justification (Why HLM?)

Problem with OLS:

- Loans are clustered within government entities, violating OLS independence assumptions.
- Leads to biased standard errors and unreliable results

ICC = 6.2%

- About 6.2% of cost variation comes from government type alone.
- This is non-trivial, it justified a multilevel structure

Why not Fixed Effects?

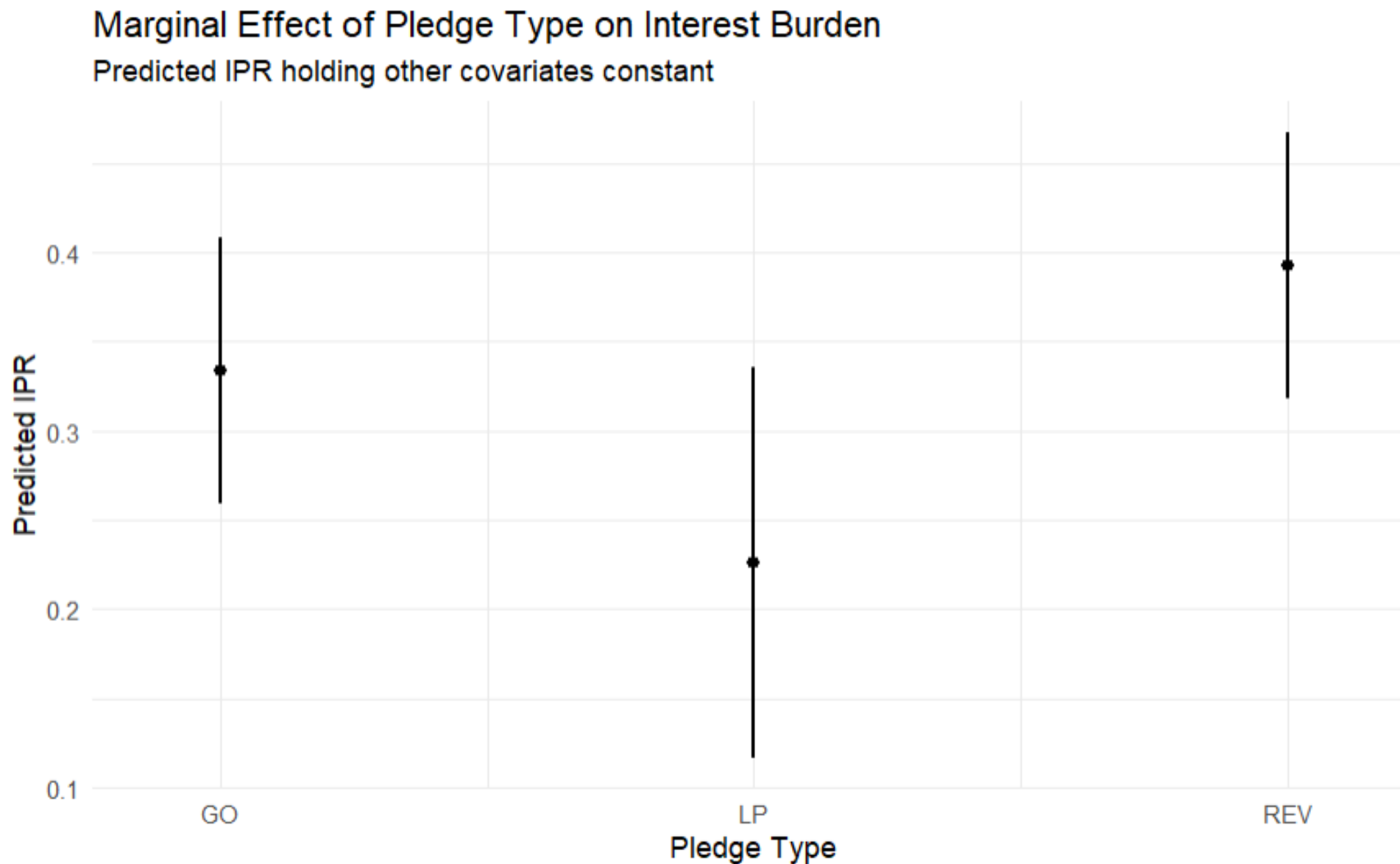
Fixed effects remove between-group variation -> we cannot generalize across entity types.

HLM Advantage (Partial Pooling)

- Balances within-group and cross-group information for stable, efficient estimates.
- Accounts for group differences without discarding them.



➔ Key Finding 1 -The Revenue Premium



The “Security Tax” of Pledge Type

Result:

Revenue (REV) bonds exhibit **higher predicted IPR** than **GO bonds**, controlling for size, purpose, and timing.

Magnitude:

REV bonds cost **~0.06 units** more than GO bonds.

Statistical Significance:

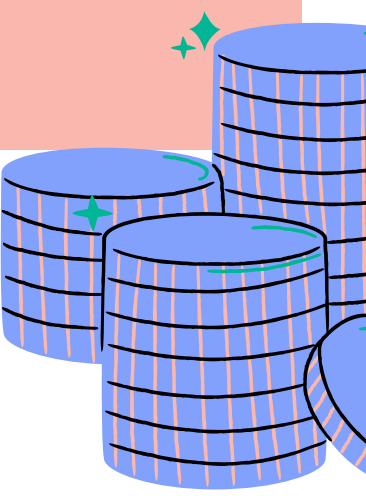
- Confidence intervals donot overlap.
- **Statistically significant difference** ($p < 0.001$)

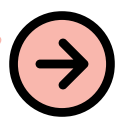
Interpretation:

Indicates a persistent **risk premium** – revenue bonds carry more risk, and lenders price it in.

Lease Purchase (LP) bonds:

Lower predicted IPR but **high uncertainty** due to small sample size.





Key Finding 2 - Institutional Winners & Losers

Result :

Different government types have different baseline borrowing costs, even after controlling for size & structure.

Winners (Lower cost IPR)

- Community Colleges (-0.07)
- Counties (-0.06)
- **Lower baseline borrowing cost**

Losers (Higher cost IPR)

- School Districts (+0.05)
- Water Districts (+0.04)
- **Higher baseline borrowing cost**

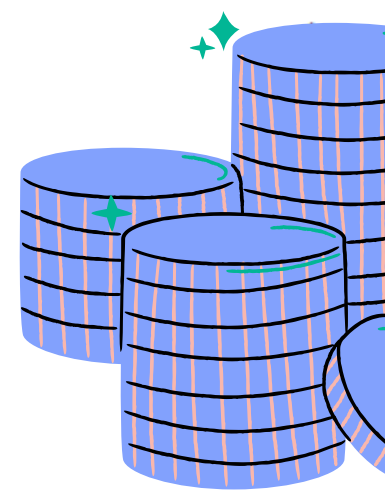
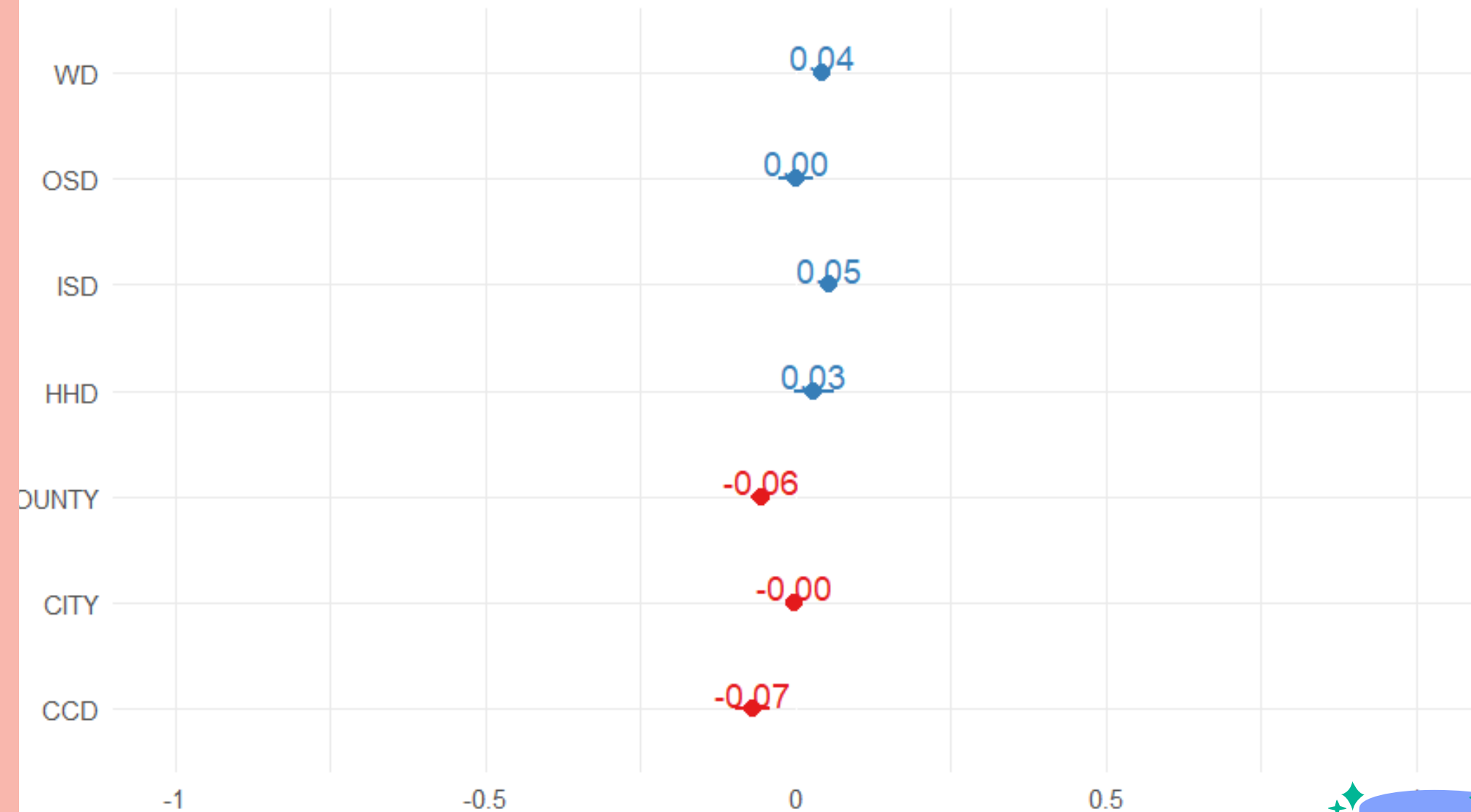
Interpretation

Borrowing cost is not just about market conditions, who you are as an institution matters.

Reliability : ICC = 6.2%

Estimates are stabilized via **partial pooling**, so results hold even for smaller groups.

Random Effects: Deviation by Government Type
Intercept shifts (u_{0j}) relative to the Grand Mean



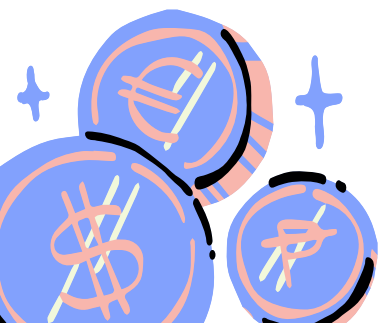
Conclusion & Future Improvements

Conclusion: The Structural Cost of Debt

- **Revenue Premium**
REV bonds carry a ~0.06 higher IPR than GO, holding other factors constant.
- **Institutional Effect**
~6.2% of variance is attributable to **entity-level heterogeneity**
Borrower identity materially impacts cost
- **Scale Effect**
Borrowing costs follow a **non – linear relationship with size**
Economies of scale with diminishing returns

Future Improvements: Refining the Model

- **Credit Risk Integration**
Include bond ratings to separate **structural vs credit-driven risk**
- **Macro Factors**
Incorporate interest rates / inflation to capture **time-varying cost dynamics**
- **Extended Hierarchy**
Add geographic layers to model **regional heterogeneity**



*Thank you
Any Questions?*

